



NETWORK INFRASTRUCTURE FOR ONLINE LEARNING PLATFORM

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2CS1

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PROJECT INFORMATION

Project Title : Network Infrastructure For Online Learning Platform

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CERTIFICATE OF ORIGINALITY

This is to certify that the project report titled "Network Infrastructure For Online Learning Platform" is an original work completed by Azzahra Fatma Wahidah, Hilwa Annisa, and Nazilatun Natasya Ramadhani. This project has been submitted in partial fulfillment of their course requirement at the National Institute of Information Technology (NIIT).

The project report has been prepared under our research and experiment, and it is ensured that the work presented in this report is the result of the individual efforts of the aforementioned students. The contents of this report have not been submitted to any other institution or organization for the award of any degree, diploma, or other similar recognition.

Authors acknowledge that the ideas, designs, and implementations presented in this project report are the intellectual properties of the students mentioned above. Any use or reproduction of this work must give proper credit to the original authors.

Authors hereby endorse the authenticity and originality of the work presented in this project report and confirm that it meets the academic standards and requirements set forth by the National Institute of Information Technology (NIIT).

Coordinator:

Mr. Ivan Firdaus

ACKNOWLEDGEMENT

The author would like to acknowledge the completion of a very useful paper entitled “Network Infrastructure for Online Learning Platforms”. This paper comprehensively discusses the integration of Network Technology in the function of analyzing efficiency, security, and convenience within the infrastructure.

The content of the paper provides a detailed overview of the potential benefits of Network Technology. The authors have carefully examined various aspects of Network Technology, such as Routing, IP Services, Web Servers, DNS Servers, FTP Servers and other Network services. In addition, the paper explores the challenges associated with the implementation of Networking at LUN8 Learning Center, offering valuable insights for future research and development in this area.

Overall, this paper makes a significant contribution to the ever-growing body of knowledge on Networking applications in the context of network infrastructure deployments.

Depok, March 6 2025

Authors

SYSTEM ANALYSIS

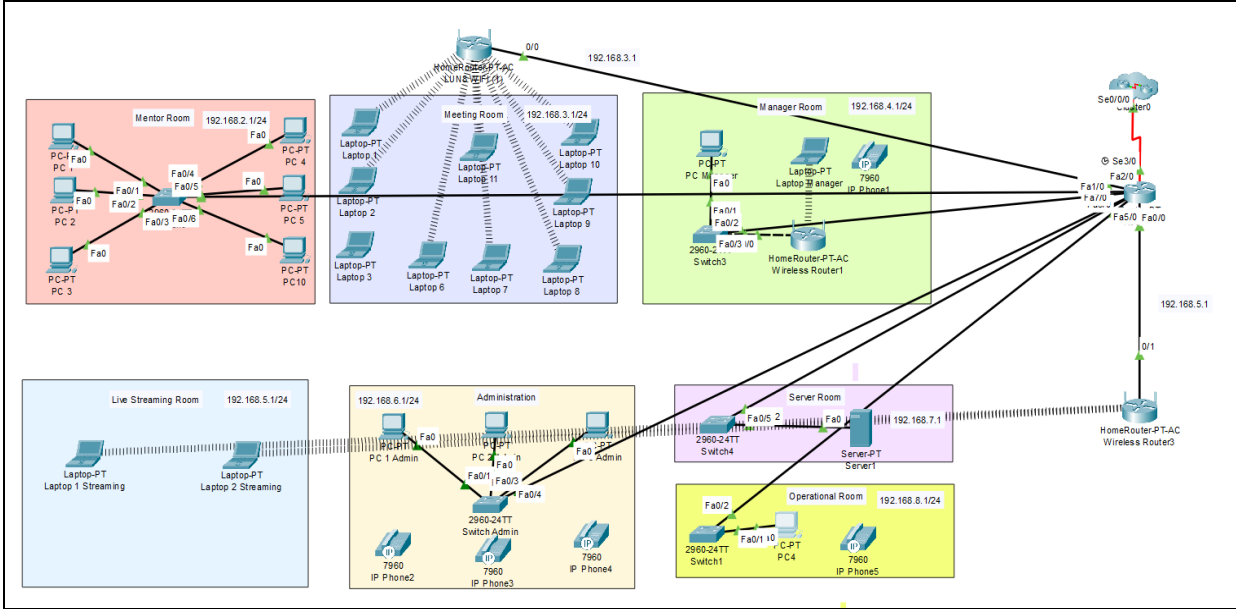
This paper reviews the implementation and development of network infrastructure in supporting online learning activities at LUN8 Learning Center. The main focus of this research is to comprehensively analyze the network architecture used, with the aim to evaluate the level of efficiency, connection stability, and scalability flexibility.

The research explored various network components, including routers, switches, cables, access points, and other supporting elements. The research was conducted to identify potential bottlenecks, security risks, and optimization opportunities to improve the performance of the existing system. In addition, this paper discusses the application of innovative technologies and best practices in network design and management, considering aspects of bandwidth requirements, cyber threat mitigation, and network readiness for future growth.

By applying empirical studies, modeling, and simulation-based approaches, this research aims to formulate effective strategies to improve network infrastructure performance at LUN8 Learning Center. These measures are designed to ensure more reliable connections, stronger security systems, as well as efficiency in resource utilization.

Ultimately, this research establishes a strategic framework for designing, developing, and maintaining a network infrastructure that meets the specific needs of LUN8 Learning Center. As such, it contributes to improving communication effectiveness, strengthening collaboration, and optimizing productivity in the academic and professional ecosystem.

NETWORK TOPOLOGY



SECTION	NETWORK ADDRESS	SUBNETMASK	STATUS
OPERATIONAL ROOM	192.168.8.1	255.255.255.0	STATIC
SERVER ROOM	192.168.7.1	255.255.255.0	STATIC
ADMINISTRATION ROOM	192.168.6.1	255.255.255.0	STATIC
LIVE STREAMING ROOM	192.168.5.1	255.255.255.0	DHCP
MANAGER ROOM	192.168.4.1	255.255.255.0	STATIC
MEETING ROOM	192.168.3.1	255.255.255.0	DHCP
MENTOR ROOM	192.168.2.1	255.255.255.0	DHCP

NETWORK DEVICES

FLOOR 1

Devices	Device Name	IP Address	Gateway	Features
Router-PT	Main Router	SE 3/0 192.168.50.1 /24 SE 0/0 192.168.50.1	ISP IP	- Routing Static - CMD
Wireless Router	LUN8 WIFI(2)	FA0/2 192.168.40.1 /24 FA0/1	-	-
Server-PT	Server	192.168.7.2	192.168.7.1	- DNS - FTP - Web Server
Switch	SwitchSV	-	-	-
Switch	SwitchOP	-	-	-
Switch	SwitchAdm	-	-	-
PC	OP PC	192.168.8.3	192.168.8.1	-Browser -CMD
PC	PC 1 Adm	192.168.6.3	192.168.6.1	- Browser - CMD
PC	PC 2 Adm	192.168.6.4	192.168.6.1	- Browser - CMD
PC	PC 3 Adm	192.168.6.5	192.168.6.1	- Browser - CMD

NETWORK DEVICES

Laptop	Lp1LS	192.168.5.3	192.168.5.1	- Browser - CMD
Laptop	Lp1LS	192.168.5.6	192.168.5.1	- Browser - CMD

FLOOR 2

Devices	Device Name	IP Address	Gateway	Features
Router-PT	Main Router	SE 3/0 192.168.50. 1/24 SE 0/0 192.168.50. 1	ISP IP	- Routing Static - CMD
Wireless Router	LUN8 WIFI(1)	FA2/0 192.168.3.1 /24	-	- Wireless
Wireless Router	WIFIManager	FA0/3 192.168.6.1 /24	-	- Wireless
Switch	Swtmanager	-	-	-
Switch	SwtMentor	-	-	-
PC	PC Manager	192.168.4.3	192.168.4.1	-Browser -CMD
PC	PC1	192.168.2.3	192.168.2.1	- Browser - CMD

NETWORK DEVICES

PC	PC2	192.168.2.4	192.168.2.3	-Browser - CMD
PC	PC3	192.168.2.5	192.168.2.1	- Browser - CMD
PC	PC4	192.168.2.6	192.168.2.3	- Browser - CMD
PC	PC5	192.168.2.7	192.168.2.1	- Browser - CMD
PC	PC6	192.168.2.8	192.168.2.1	- Browser - CMD
Laptop	Laptop 1	191.168.3.3	191.168.3.1	- Browser - CMD
Laptop	Laptop 2	191.168.3.4	191.168.3.1	- Browser - CMD
Laptop	Laptop 3	191.168.3.5	191.168.3.1	- Browser - CMD
Laptop	Laptop 4	191.168.3.6	191.168.3.1	- Browser - CMD

NETWORK DEVICES

				-
Laptop	Laptop 6	191.168.3.8	191.168.3.1	- Browser - CMD
Laptop	Laptop 7	191.168.3.9	191.168.3.1	- Browser - CMD
Laptop	Laptop 8	191.168.3.10	191.168.3.1	- Browser - CMD
Laptop	Laptop 9	191.168.3.11	191.168.3.1	- Browser - CMD

NETWORK DEVICES

Router IP 1 Configuration

mainrouter

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet0/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps

☐ 10 Mbps

Duplex

☐ Half Duplex

☒ Full Duplex

MAC Address0006.2A71.920C

IP Configuration

IPv4 Address192.168.5.1

Subnet Mask255.255.255.0

Tx Ring Limit10

Router IP 2 Configuration

mainrouter

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet1/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps

☐ 10 Mbps

Duplex

☐ Half Duplex

☒ Full Duplex

MAC Address00D0.972C.CA00

IP Configuration

IPv4 Address192.168.4.1

Subnet Mask255.255.255.0

Tx Ring Limit10

NETWORK DEVICES

Router IP 3 Configuration

mainrouter

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet2/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0.BAAB.8854

IP Configuration

IPv4 Address 192.168.3.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Router IP 4 Configuration

mainrouter

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

Serial3/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 192.168.50.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

NETWORK DEVICES

Router IP 5 Configuration

 mainrouter

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet4/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.4204.091A

IP Configuration

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Router IP 6 Configuration

 mainrouter

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet5/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.A3CE.4932

IP Configuration

IPv4 Address 192.168.7.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

NETWORK DEVICES

Router IP 7 Configuration

mainrouter

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet6/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps ☐ 10 Mbps

Duplex

☐ Half Duplex ☒ Full Duplex

MAC Address000C.CF29.E548

IP Configuration

IPv4 Address192.168.8.1

Subnet Mask255.255.255.0

Tx Ring Limit10

Router IP 8 Configuration

mainrouter

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

FastEthernet7/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps ☐ 10 Mbps

Duplex

☐ Half Duplex ☒ Full Duplex

MAC Address00D0.BCDA.9B6E

IP Configuration

IPv4 Address192.168.6.1

Subnet Mask255.255.255.0

Tx Ring Limit10

NETWORK DEVICES

Router Routing RIP

 mainrouter — □ ×

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

FastEthernet2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet6/0

FastEthernet7/0

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.7.0/24 via 192.168.30.1

192.168.30.0/24 via 192.168.7.0

192.168.7.0/24 via 192.168.40.1

192.168.40.0/24 via 192.168.7.0

192.168.7.0/24 via 192.168.60.1

192.168.60.0/24 via 192.168.7.0

Remove

Router Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.20.1
% Incomplete command.
Router(config-if)#ip add 192.168.20.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int se3/0
Router(config-if)#ip add 192.168.50.1 255.255.255.0
Router(config-if)#no sh
```

NETWORK DEVICES

```
Router(config-if)#int fa7/0
Router(config-if)#ip add 192.168.6.1 255.255.255.0
Router(config-if)#no sh
```

```
Router(config-if)#int fa6/0
Router(config-if)#ip add 192.168.8.1 255.255.255.0
Router(config-if)#
Router(config-if)#no sh
```

```
Router(config-if)#int fa5/0
Router(config-if)#ip add 192.168.7.1 255.255.255.0
Router(config-if)#no sh
```

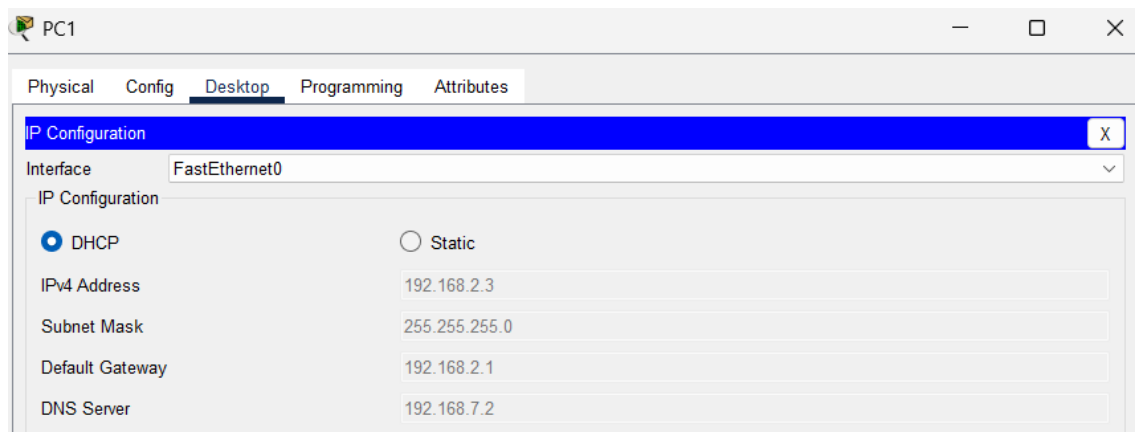
```
Router(config-if)#int fa4/0
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#no sh
```

```
Router(config-if)#int fa2/0
Router(config-if)#ip add 192.168.40.1 255.255.255.0
Router(config-if)#no sh
```

```
Router(config-if)#int fa1/0
Router(config-if)#ip add 192.168.30.1 255.255.255.0
Router(config-if)#no sh
```


FLOOR 2 CONFIGURATION

PC 1 DHCP IP Configuration



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

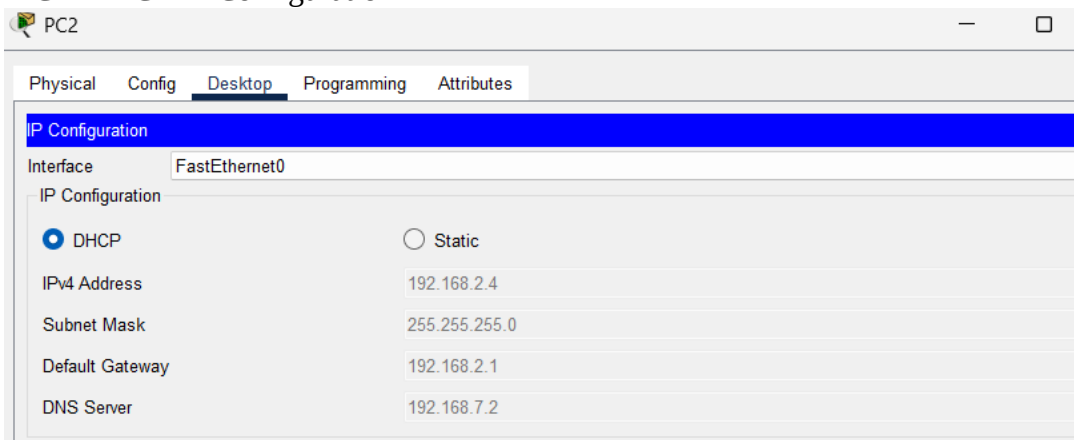
IPv4 Address 192.168.2.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.7.2

PC 2 DHCP IP Configuration



PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

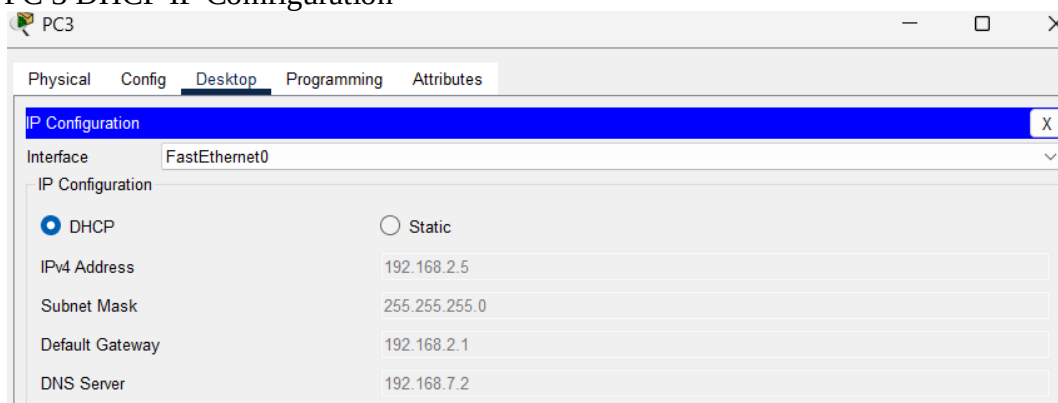
IPv4 Address 192.168.2.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.7.2

PC 3 DHCP IP Configuration



PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.2.5

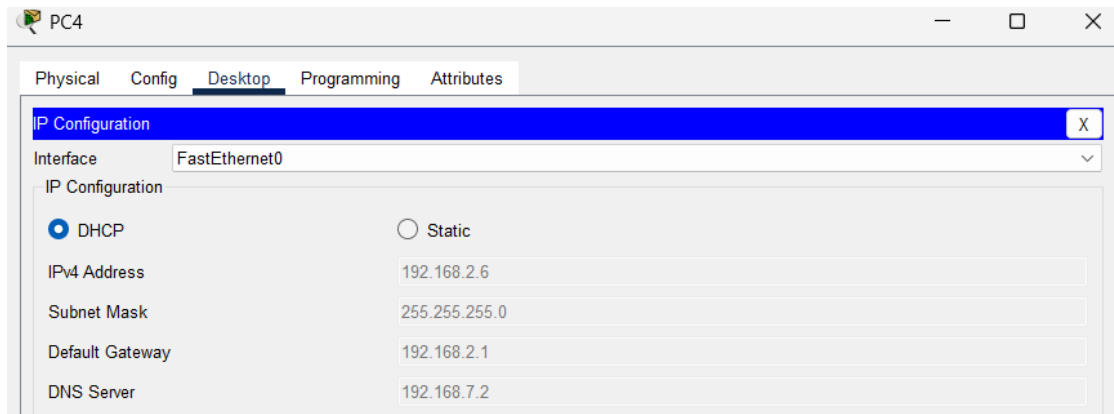
Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.7.2

FLOOR 2 CONFIGURATION

PC 4 DHCP IP Configuration



PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

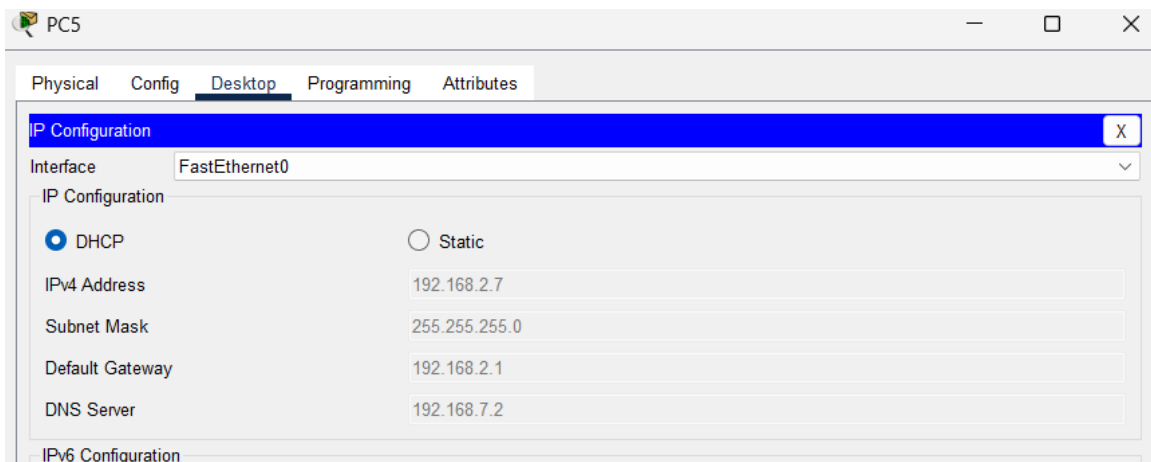
IPv4 Address 192.168.2.6

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.7.2

PC 5 DHCP IP Configuration



PC5

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.2.7

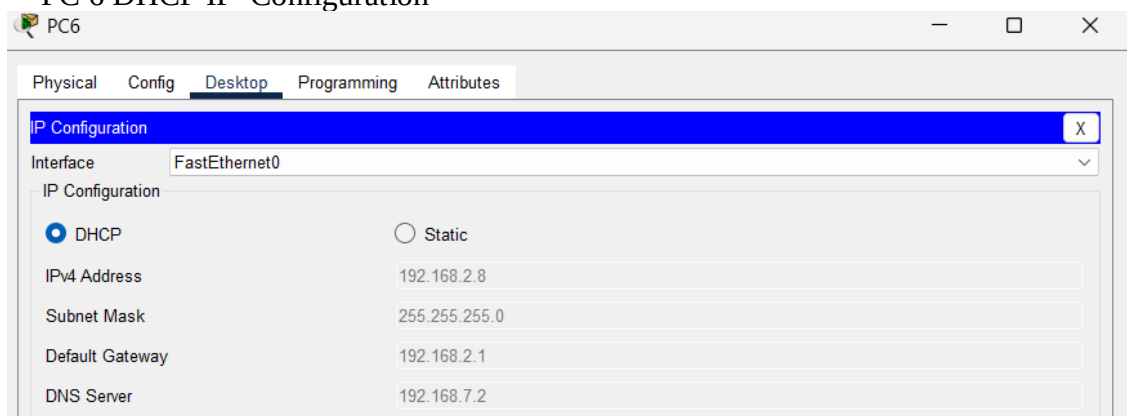
Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 192.168.7.2

IPv6 Configuration

PC 6 DHCP IP Configuration



PC6

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.2.8

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

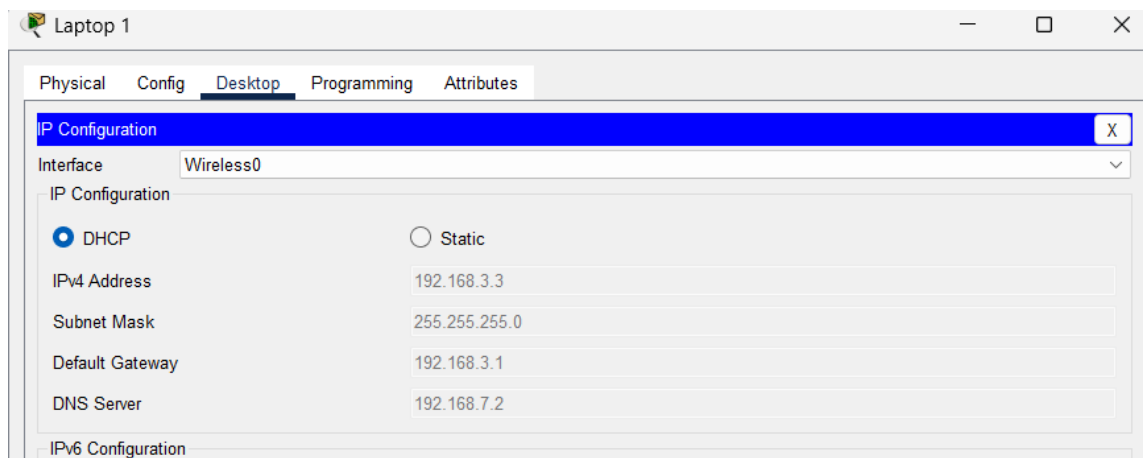
DNS Server 192.168.7.2

FLOOR 2 CONFIGURATION

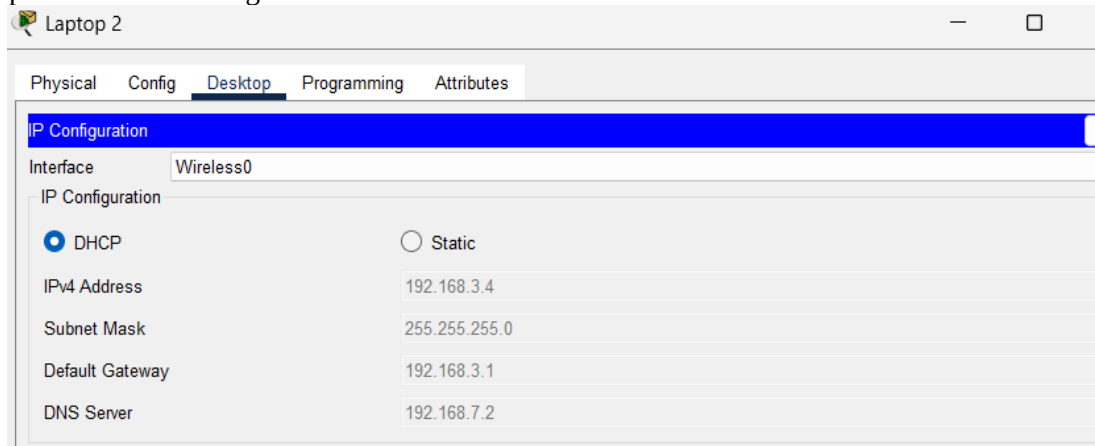
CLI Turning IP 2 to DHCP

```
Router(config)#ip dhcp pool mentorroom
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.80.1
Router(dhcp-config)#dns-server 192.168.90.9
Router(dhcp-config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Laptop 1 DHCP IP Configuration

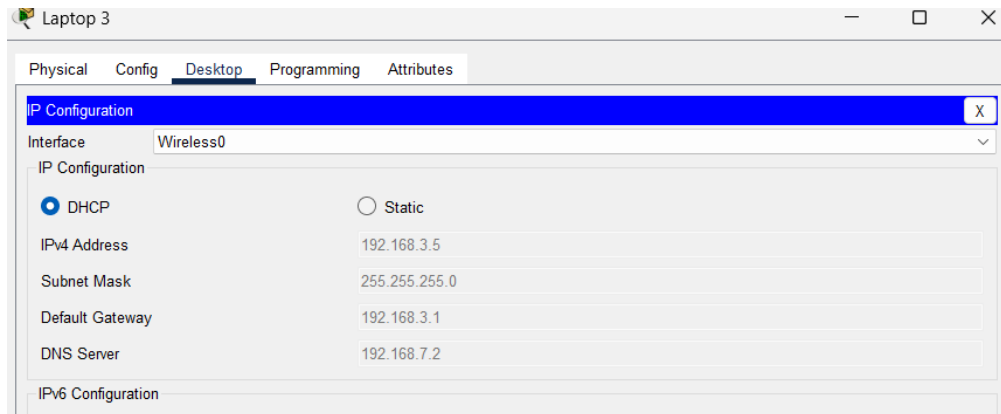


Laptop 2 DHCP IP Configuration



FLOOR 2 CONFIGURATION

Laptop 3 DHCP IP Configuration

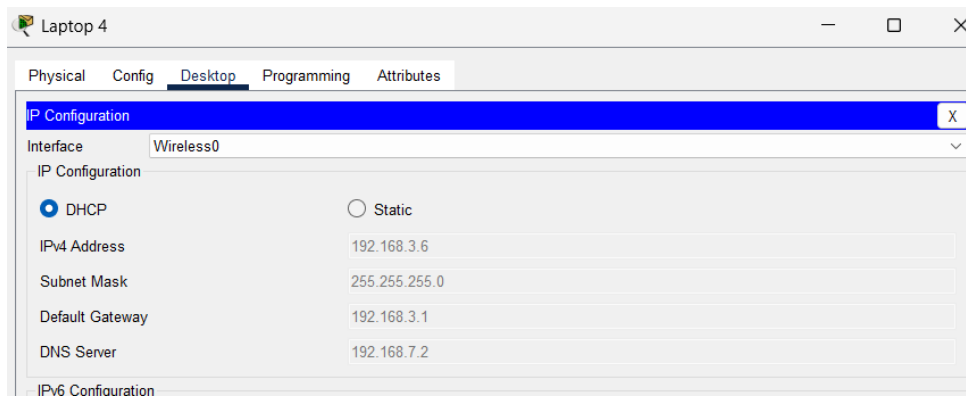


The screenshot shows the 'Laptop 3' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'Wireless0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.5
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

The 'IPv6 Configuration' section is collapsed.

Laptop 4 DHCP IP Configuration

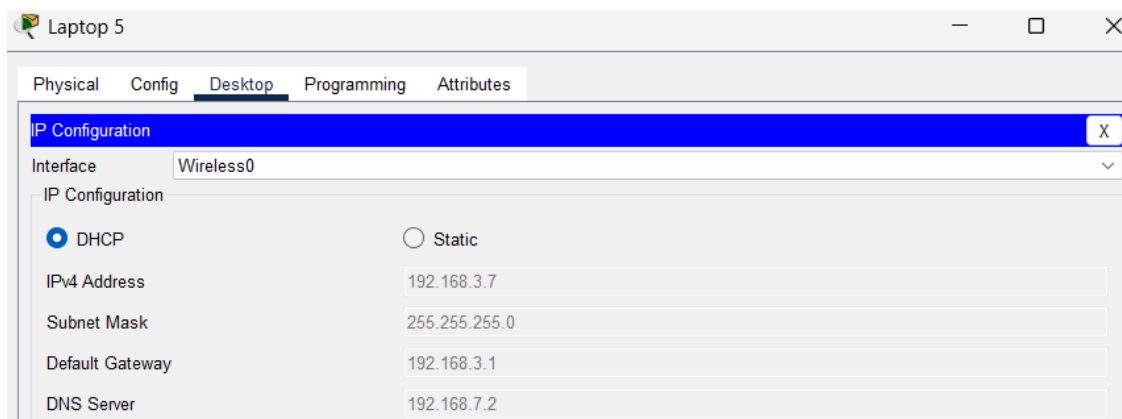


The screenshot shows the 'Laptop 4' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'Wireless0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.6
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

The 'IPv6 Configuration' section is collapsed.

Laptop 5 DHCP IP Configuration



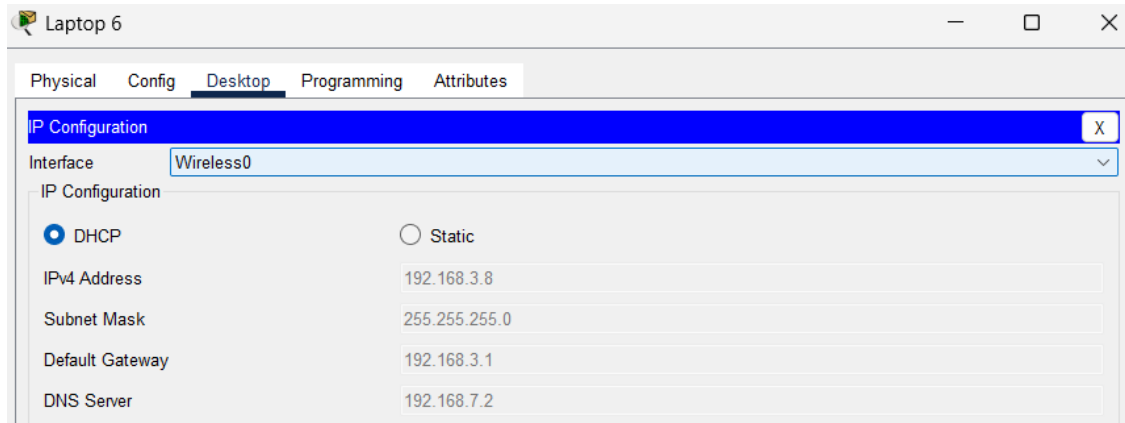
The screenshot shows the 'Laptop 5' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'Wireless0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.7
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

The 'IPv6 Configuration' section is collapsed.

FLOOR 2 CONFIGURATION

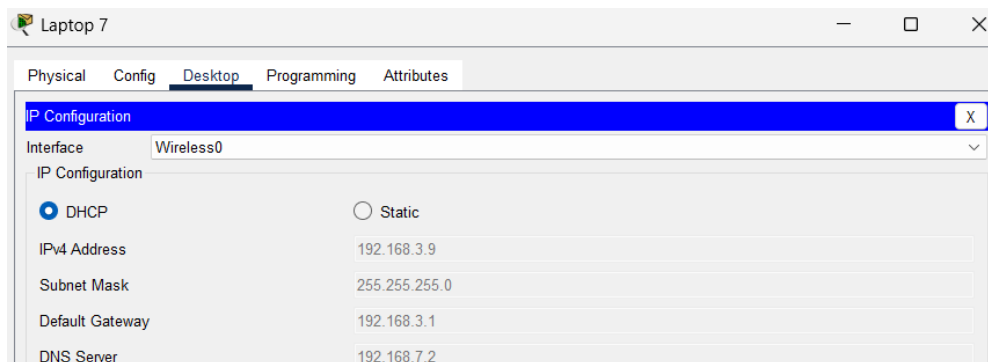
Laptop 6 DHCP IP Configuration



The screenshot shows the 'Laptop 6' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'Wireless0'. The 'IP Configuration' section shows 'DHCP' selected with a radio button, and 'Static' is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.8
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

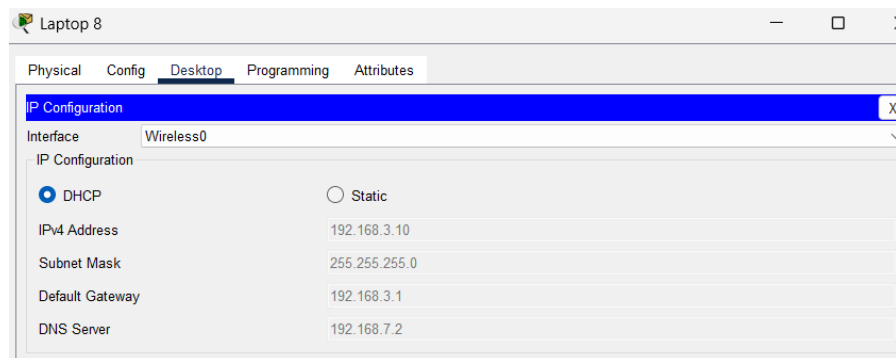
Laptop 7 DHCP IP Configuration



The screenshot shows the 'Laptop 7' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'Wireless0'. The 'IP Configuration' section shows 'DHCP' selected with a radio button, and 'Static' is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.9
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

Laptop 8 DHCP IP Configuration

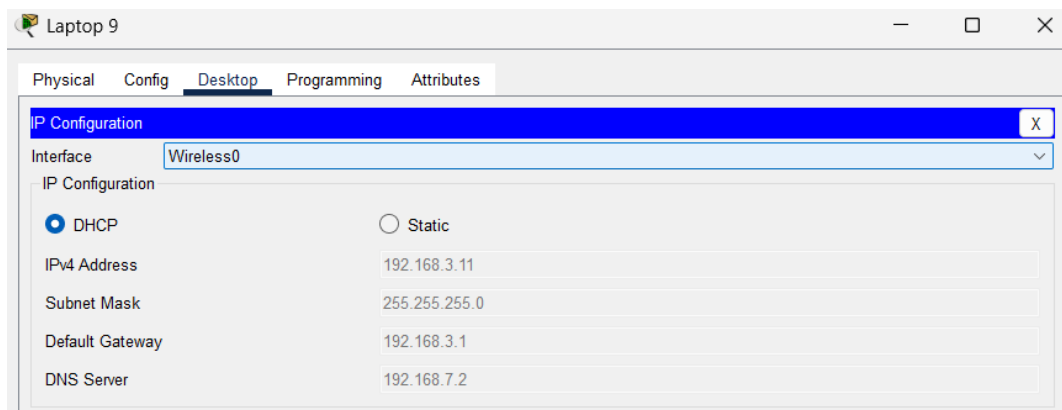


The screenshot shows the 'Laptop 8' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'Wireless0'. The 'IP Configuration' section shows 'DHCP' selected with a radio button, and 'Static' is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

FLOOR 2 CONFIGURATION

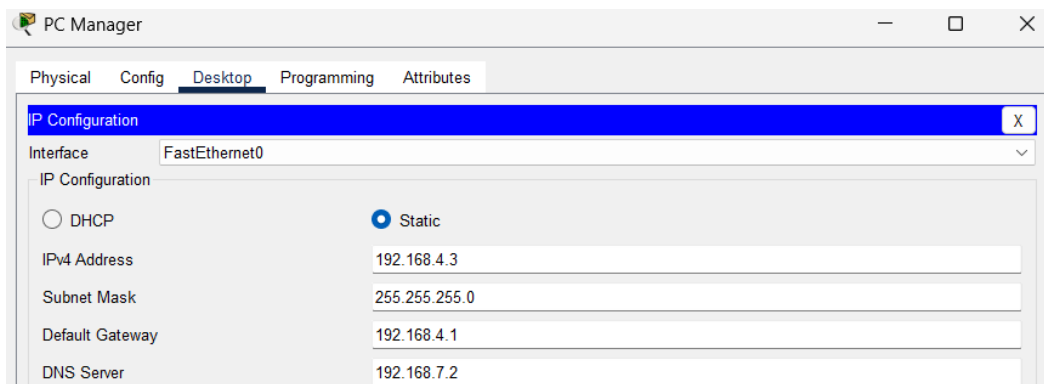
Laptop 9 DHCP IP Configuration



The screenshot shows the 'Laptop 9' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing the 'Wireless0' interface. The 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.3.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.3.1
DNS Server	192.168.7.2

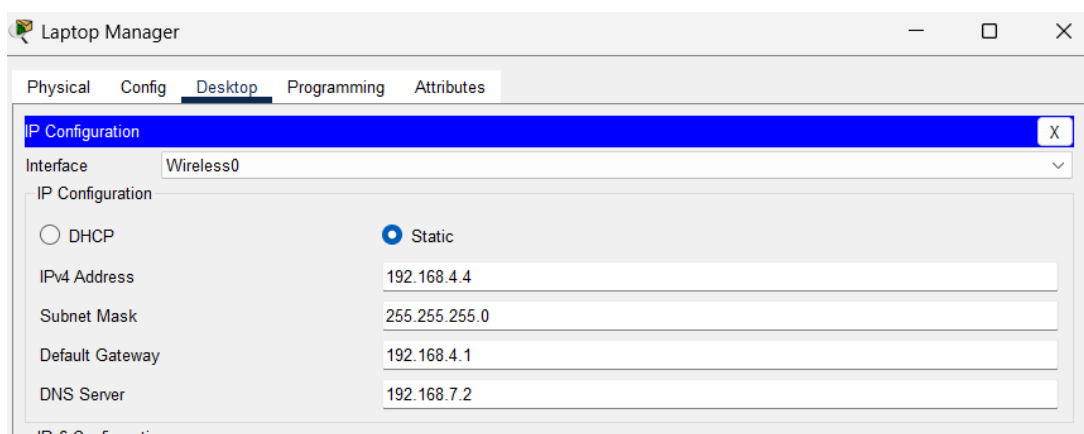
PC Manager Static IP Configuration



The screenshot shows the 'PC Manager' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing the 'FastEthernet0' interface. The 'Static' radio button is selected, and the 'DHCP' radio button is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.4.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	192.168.7.2

Laptop Manager Static IP Configuration

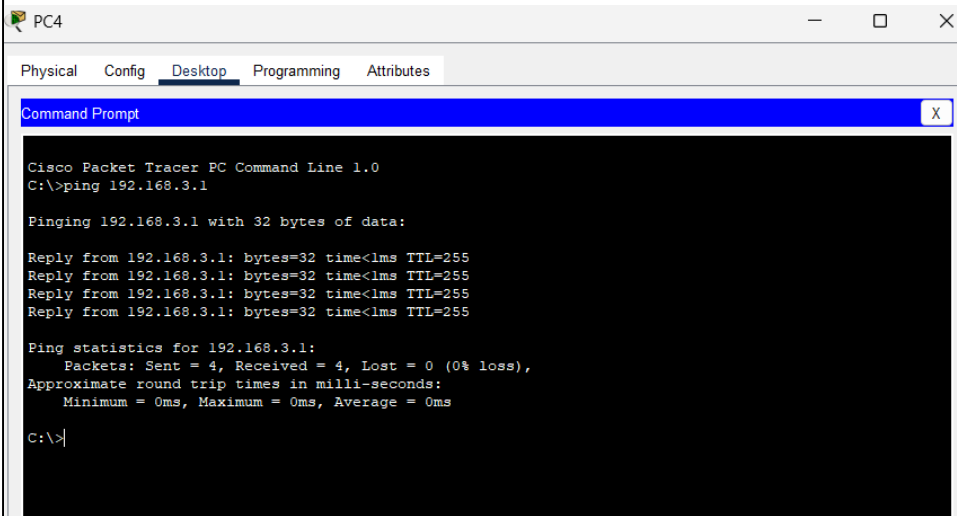


The screenshot shows the 'Laptop Manager' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing the 'Wireless0' interface. The 'Static' radio button is selected, and the 'DHCP' radio button is unselected. The fields for IPv4 Address, Subnet Mask, Default Gateway, and DNS Server are populated with the following values:

Field	Value
IPv4 Address	192.168.4.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	192.168.7.2

FLOOR 2 SIMULATION

Ping PC Mentor Room to Meeting Room



The screenshot shows a Cisco Packet Tracer PC window for PC4. The 'Desktop' tab is active, displaying a Command Prompt. The user has entered the command 'ping 192.168.3.1'. The output shows four successful replies from 192.168.3.1 with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The ping statistics indicate that all four packets were sent and received with 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1

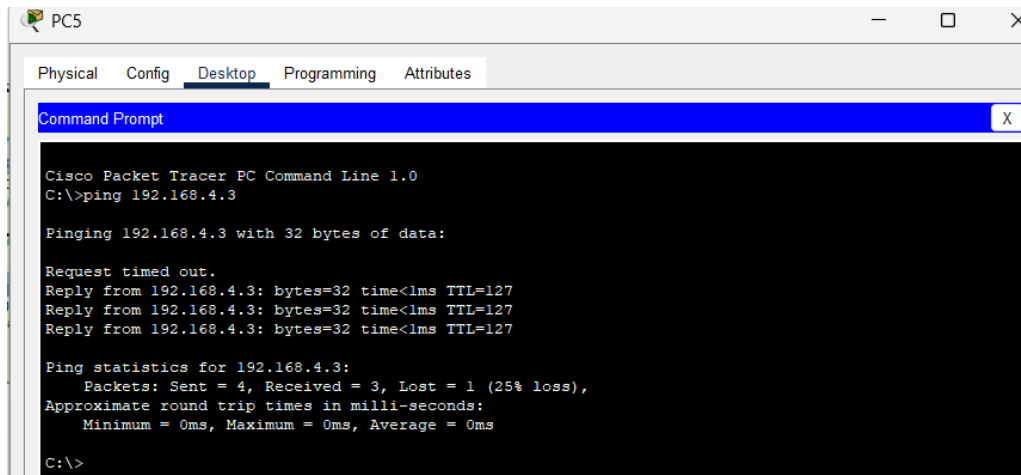
Pinging 192.168.3.1 with 32 bytes of data:

Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255
Reply from 192.168.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Ping PC SE Class to WD



The screenshot shows a Cisco Packet Tracer PC window for PC5. The 'Desktop' tab is active, displaying a Command Prompt. The user has entered the command 'ping 192.168.4.3'. The output shows a 'Request timed out' message, followed by three successful replies from 192.168.4.3 with 32 bytes of data, a time of less than 1ms, and a TTL of 127. The ping statistics indicate that one packet was lost out of four sent, resulting in a 25% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.3

Pinging 192.168.4.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.3: bytes=32 time<1ms TTL=127
Reply from 192.168.4.3: bytes=32 time<1ms TTL=127
Reply from 192.168.4.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.4.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

FLOOR 2 SIMULATION

Ping SE Class to Admin Room

```
C:\>ping 191.168.1.2

Pinging 191.168.1.2 with 32 bytes of data:

Reply from 191.168.1.2: bytes=32 time=4ms TTL=125
Reply from 191.168.1.2: bytes=32 time=4ms TTL=125
Reply from 191.168.1.2: bytes=32 time=13ms TTL=125
Reply from 191.168.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 191.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 13ms, Average = 5ms

C:\>
```

Ping SE Class to Chill Room

```
C:\>ping 1.0.0.3

Pinging 1.0.0.3 with 32 bytes of data:

Reply from 1.0.0.3: bytes=32 time=28ms TTL=128
Reply from 1.0.0.3: bytes=32 time=17ms TTL=128
Reply from 1.0.0.3: bytes=32 time=26ms TTL=128
Reply from 1.0.0.3: bytes=32 time=7ms TTL=128

Ping statistics for 1.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 28ms, Average = 19ms

C:\>
```


FLOOR 1 CONFIGURATION

Router IP Port FastEthernet0/0 Configuration

Physical	Config	CLI	Attributes
GLOBAL			
Settings			
Algorithm Settings			
ROUTING			
Static			
RIP			
INTERFACE			
FastEthernet0/0			
FastEthernet1/0			
FastEthernet2/0			
Serial3/0			
FastEthernet4/0			
FastEthernet5/0			
FastEthernet6/0			
FastEthernet7/0			

FastEthernet0/0	
Port Status	☒ On
Bandwidth	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0006.2A71.920C
IP Configuration	
IPv4 Address	192.168.5.1
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Router IP Port FastEthernet5/0 Configuration

Physical	Config	CLI	Attributes
GLOBAL			
Settings			
Algorithm Settings			
ROUTING			
Static			
RIP			
INTERFACE			
FastEthernet0/0			
FastEthernet1/0			
FastEthernet2/0			
Serial3/0			
FastEthernet4/0			
FastEthernet5/0			
FastEthernet6/0			
FastEthernet7/0			

FastEthernet5/0	
Port Status	☒ On
Bandwidth	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	00E0.A3CE.4932
IP Configuration	
IPv4 Address	192.168.7.1
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Router IP Port FastEthernet7/0 Configuration

Physical	Config	CLI	Attributes
GLOBAL			
Settings			
Algorithm Settings			
ROUTING			
Static			
RIP			
INTERFACE			
FastEthernet0/0			
FastEthernet1/0			
FastEthernet2/0			
Serial3/0			
FastEthernet4/0			
FastEthernet5/0			
FastEthernet6/0			
FastEthernet7/0			

FastEthernet7/0	
Port Status	☒ On
Bandwidth	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	00D0.BCDA.9B6E
IP Configuration	
IPv4 Address	192.168.6.1
Subnet Mask	255.255.255.0
Tx Ring Limit	10

FLOOR 1 CONFIGURATION

PC 1 Admin Static IP Configuration

Physical Config Desktop Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.6.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.6.1

DNS Server: 192.168.7.2

IPv6 Configuration

PC 2 Admin Static IP Configuration

Physical Config Desktop Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.6.4

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.6.1

DNS Server: 192.168.7.2

Lp1LS IP Configuration

Physical Config Desktop Programming Attributes

IP Configuration [X]

Interface: Wireless0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.5.6

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.5.1

DNS Server: 192.168.7.2

FLOOR 1 CONFIGURATION

Lp2LS IP Configuration

Physical Config Desktop Programming Attributes

IP Configuration X

Interface Wireless0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.5.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.5.1

DNS Server 192.168.7.2

OP PC Static IP Configuration

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.8.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.8.1

DNS Server 192.168.7.2

Server Static IP Address Configuration

Physical Config Services Desktop Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.7.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.7.1

DNS Server 192.168.7.2

FLOOR 1 CONFIGURATION

Home Router PT-AC Configuration

Physical Config **GUI** Attributes

Wireless Tri-Band Home Router Firmware V

Setup Setup Wireless Security Access Restrictions Applications & Gaming Administration Status
Basic Setup DDNS MAC Address Clone Advanced Routing

Internet Setup

Internet Connection type Automatic Configuration - DHCP Help...

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP

IP Address: 192 . 168 . 5 . 1

Subnet Mask: 255.255.255.0

DHCP Server Settings

DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.5. 3

Maximum number of Users: 50

IP Address Range: 192.168.5. 3 - 52

Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 192 . 168 . 7 . 2

Static DNS 2: 0 . 0 . 0 . 0

Static DNS 3: 0 . 0 . 0 . 0

FLOOR 2 SIMULATION

Ping from Laptop 1 Live Streaming PC to Admin Room PC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 192.168.6.3

Pinging 192.168.6.3 with 32 bytes of data:

Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.

Ping statistics for 192.168.6.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Ping from Laptop 1 Live Streaming PC to Server Room PC

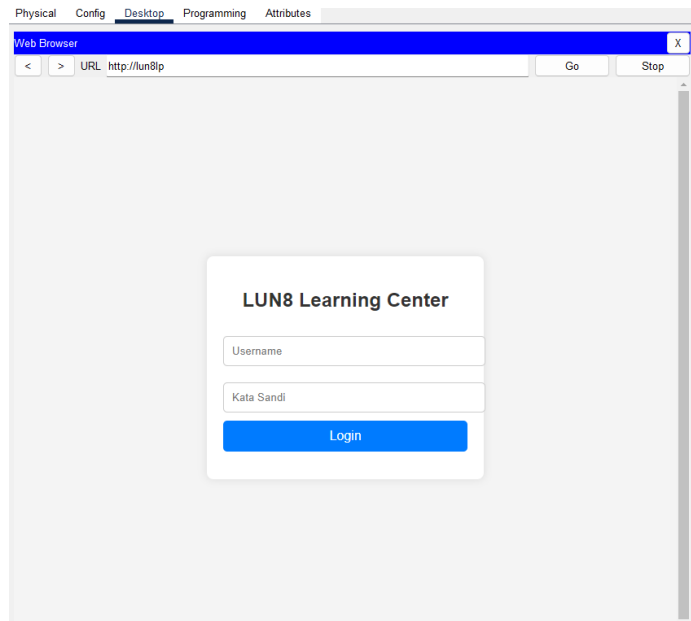
```
C:\>ping 192.168.8.3

Pinging 192.168.8.3 with 32 bytes of data:

Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.

Ping statistics for 192.168.8.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

DNS Server Test



REQUIREMENTS

Hardware :

1. Laptop HP intel core i5

Operating System :

1. Windows 11 64-bit

Software :

1. Cisco Packet Tracer
2. Ms. Word
3. Google Drive

PROJECT FILE DETAILS

No	Filename	Remarks
1	Grup 10 Project 1.pdf	Paper File
2	Project 1.pkt	Packet Tracer File
3	Project 1 Presentation.pptx	Presentation